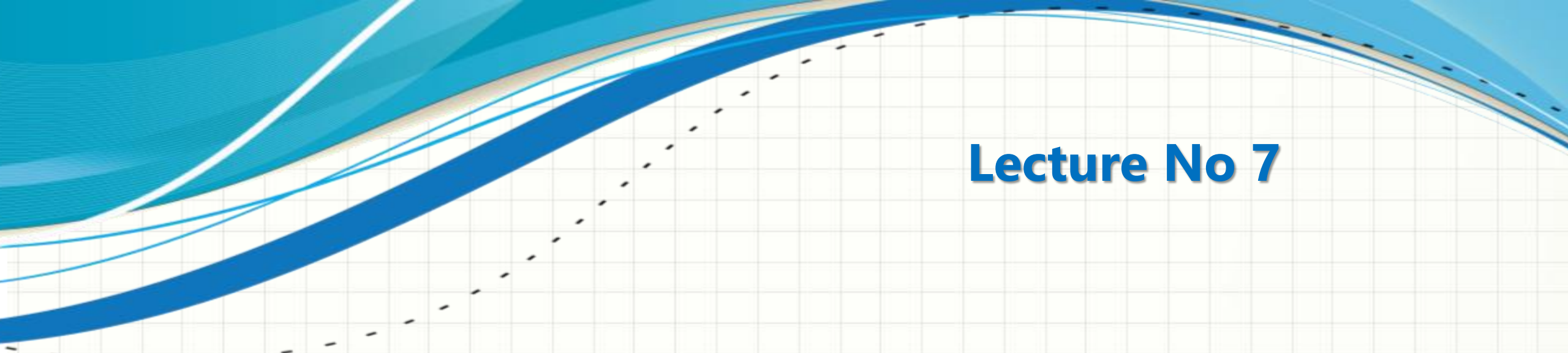




CSC432 – INFORMATION SECURITY

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Lecture No 7

RIVEST CIPHER 4

Origins of Rivest Cipher 4



- ❑ A symmetric key stream cipher designed in 1987 by Ron Rivest
- ❑ It was kept as a trade secret by RSA Security
- ❑ In September 1994, the Rivest Cipher 4 (RC4) algorithm was anonymously posted on the Internet
- ❑ Variable key-size stream cipher with byte-oriented operations
 - ❑ variable size key: 1 to 256 bytes (8 to 2048 bits)
- ❑ Currently used with a key of 128 bits
- ❑ Used in secure web browsing (SSL/TLS) and wireless LANs (WEP)
- ❑ Simple and efficient implementation



The RC4 Algorithm

Parameters and Variables

- ❑ Variable length key, K , from 1 to 256 Bytes
- ❑ State vector, S , 256 Bytes
 - ❑ S contains a permutation of all 8-bit numbers from 0 through 255
- ❑ Temporary vector, T , 256 Bytes
- ❑ A byte from keystream, k , generated from S

Steps

- ❑ Initialize S to values 0 to 255; initialize T with repeating values of key, K
- ❑ Use T to create initial permutation of S
- ❑ Permute S and generate keystream, k from S
- ❑ Encrypt a byte of plaintext, p , by XOR with k

$$T[i] = K[i \bmod \text{keylen}];$$


RC4



The RC4 Algorithm

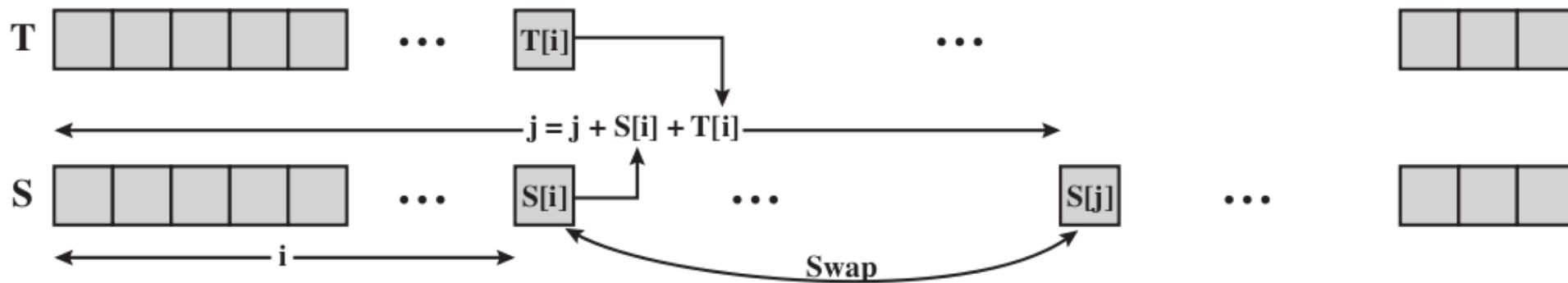
Initial Permutation of S (also known as Key Scheduling Algorithm)

$j = 0;$

for $i = 0$ to 255 do

$j = (j + S[i] + T[i]) \bmod 256;$

Swap ($S[i], S[j]$);



RC4



The RC4 Algorithm

Stream Generation (also known as Pseudo Random Number Generation (PRNG))

$i, j = 0;$

while (true)

$i = (i + 1) \bmod 256;$

$j = (j + S[i]) \bmod 256;$

Swap ($S[i], S[j]$);

$t = (S[i] + S[j]) \bmod 256;$

$k = S[t];$

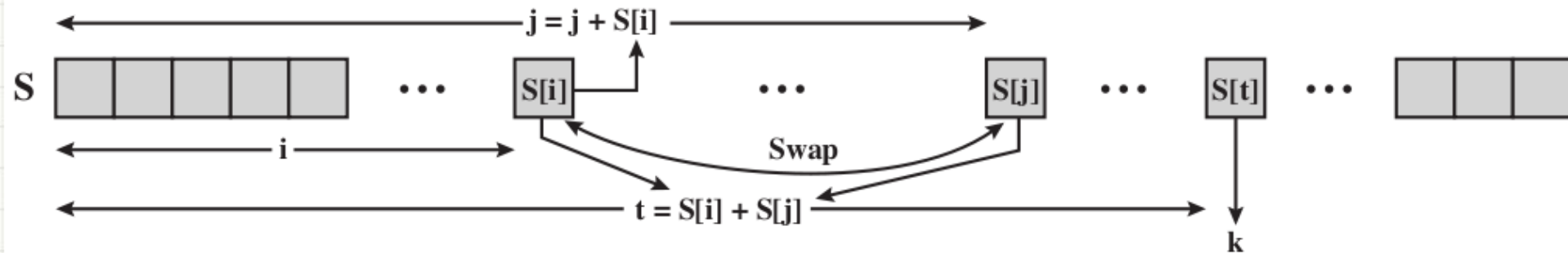
* A pseudorandom number is one that is generated by an algorithm but is unpredictable without knowledge of the input key

RC4



The RC4 Algorithm

Stream Generation



To encrypt: $C = P \text{ XOR } k$

To decrypt: $P = C \text{ XOR } k$

RC4



Strengths

- ☐ Difficulty of knowing the state
- ☐ Difficulty of knowing which location in the state table is used to select each value in the sequence
- ☐ Encryption in software is about 10 times faster than DES



Weaknesses

- ☐ Vulnerable to analytic attacks to the state table
- ☐ Theoretical attacks performed on the RC4 algorithm; one of the best known has been developed by Fhlurer, Martin and Shamir
- ☐ There are some weak keys, so the initial 512 bytes of output is dropped in order to defeat weak key attacks and other attacks related to key schedule
- ☐ None of these approaches is practical against RC4 with a reasonable key length, such as 128 bits

Block vs Stream Ciphers



- ❑ With a properly designed PRNG, a stream cipher can be as secure as block cipher of comparable key length
- ❑ Stream ciphers are almost always faster and use far less code than block ciphers
- ❑ Block ciphers can reuse keys
- ❑ However, if two plaintexts are encrypted with the same key using a stream cipher, then cryptanalysis is often quite simple
 - ❑ If the two ciphertext streams are XORed together, the result is the XOR of the original plaintexts
 - ❑ If the plaintexts are text strings, credit card numbers, or other byte streams with known properties, then cryptanalysis becomes more easy

Block vs Stream Ciphers



- ❑ For applications that require encryption/decryption of a stream of data, such as over a data communications channel or a browser/Web link, a stream cipher might be the better alternative
- ❑ For applications that deal with blocks of data, such as file transfer, e-mail, and database, block ciphers may be more appropriate
- ❑ However, either type of cipher can be used in virtually any application

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- ❑ RSA Security

<http://www.emc.com/domains/rsa/index.htm>

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- ❑ Weakness in the KSA

Fluhrer, S.; Mantin, I.; and Shamir, A. "Weakness in the Key Scheduling Algorithm of RC4" Proceedings, Workshop in Selected Areas of Cryptography, 2001

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- ❑ "Network Security: Private Communication in a Public World", by Charlie Kaufman, Radia Perlman, Mike Speciner (chapter 3)



THANKS